

Writing the methods section

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Figures that preserve evidence

Without opening the old lesson, explain one key idea from Figures that preserve evidence. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Robustness across conditions

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reproducible environment and records

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Writing the methods section. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Writing the methods section. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Methods says used AI on data and got good scores; code and split are absent. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A detailed methods section can reveal limitations; it does not guarantee a correct design. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled writing the methods section evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of writing the methods section. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Writing the methods section

Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.

Inputs: "oak", "pine", "willow". Rule: length at least 4→long, otherwise short. No whitespace trimming occurs.

Writing the methods section

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record exact inputs

Preserve the three strings and no-trimming policy.



2. State procedure

Apply the inclusive length boundary at four characters.



3. Show resulting labels

Oak is short; pine and willow are long.

CHANGE ONE THING

Change the rule to length greater than 4.

Explain the changed result using the matching lesson.

Apply the inclusive four-character rule to oak, pine and willow. Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Oak is short; pine and willow are long.

1. State procedure
2. Record exact inputs
3. Show resulting labels

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Show resulting labels Oak is short; pine and willow are long. A methods section specifies inputs, processing, boundaries, versions and scoring. Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

Practice 2: explain the deciding evidence

Changed-case practice: Change the rule to length greater than 4. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Pine becomes short; willow stays long and oak stays short.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Pine becomes short; willow stays long and oak stays short. The equality case distinguishes the two procedures.

Repair a missing building block

Review Figures that preserve evidence (ai-7/29). Goal: Explain the mechanism and terminology of figures that preserve evidence. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Review Robustness across conditions (ai-7/27). Goal: Explain the mechanism and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the mechanism and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the output so another team can identify the exact run?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

